

KeraNova Research Methodology

Explainable prediction of 12-month corrected distance vision

KeraNova Research Team | Version 1.0 | 1 October 2026

1 Study question and intended use

Can baseline clinical and corneal tomographic characteristics predict corrected distance visual acuity (CDVA), measured in logMAR, approximately 12 months after topography-guided photorefractive keratectomy plus accelerated corneal cross-linking (TG-PRK + ACXL) or accelerated CXL alone?

KeraNova is an exploratory research prototype. Its primary aim is to develop and internally evaluate a transparent prognostic model. Secondary aims are to describe one-year visual, refractive, keratometric and recorded safety outcomes in the two observed treatment cohorts. The model estimates outcomes under treatments actually received; it does not estimate individual causal treatment effects or recommend a procedure.

2 Design, setting and study population

The design is an exclusively retrospective, single-centre observational cohort at the Anterior Segment Clinic, King Khaled Eye Specialist Hospital (KKESH), Riyadh, Saudi Arabia. The parent protocol identifies procedures performed from January 2020 through January 2025. The proposed extraction interprets this as 01 January 2020–31 January 2025 inclusive; the investigators must confirm the exact endpoints against the approved protocol before extraction is locked.

The parent study plans up to 60 eligible eyes, approximately 30 per treatment cohort, from available records. This is a planned convenience sample, not an achieved sample or a demonstration of adequate prediction-model sample size. Enumerate the full eligible source population and record the selection mechanism. If more eyes qualify than the authorized sample permits, agree and record an outcome-blinded selection rule before extraction; do not select eyes to balance outcomes or model performance.

Requirement	Operational definition / evidence
Age	At least 18 years at the index procedure; calculate from verified dates or use age documented at treatment.
Diagnosis and progression	Clinician-confirmed progressive keratoconus in the source record. Record the clinical evidence and adjudicator. A diagnosis label or severity stage alone does not establish progression.
Permitted treatment	Operative record verifies TG-PRK + ACXL or ACXL alone, including topography guidance and accelerated CXL parameters where applicable.
Corneal safety / contraindications	Adequate thickness and absence of ocular/systemic contraindications as judged in the treatment record; do not create a new universal thickness threshold.
Exclusions	Apply the approved parent criteria: previous corneal surgery, including keratoplasty or laser vision correction; other ocular disease affecting outcomes; uncontrolled diabetes (HbA1c >8.0% or recent ketoacidosis); active autoimmune connective-tissue disease associated with corneal melting; and pregnancy/lactation or planned pregnancy where required by the actual IRB determination.

Requirement	Operational definition / evidence
Other procedures	PTK + CXL and unverified PRK/CXL variants are outside the prespecified primary cohorts; retain them in the screening log with the reason.
Insufficient evidence	Set eligibility to pending verification, rather than infer eligible or ineligible from a blank CSV field.

3 Unit of observation and chronology

The observation is an eye at its first eligible index treatment. A coded patient identifier links right and left eyes; an eye identifier and laterality identify the eye. Both eligible eyes may be retained if authorized, with patient-level clustering in all resampling and inference. Never infer patient linkage from age, sex or consecutive rows. Record any later retreatment or additional corneal procedure rather than treating it as a second independent baseline record.

Baseline is the closest valid, documented examination before the index procedure. Store its date, the interval to surgery, visual-acuity correction type and measurement quality. A measurement obtained on the procedure date is baseline only if the source establishes that it preceded treatment. Distant baseline examinations are flagged for clinical review and a prespecified sensitivity analysis; no undocumented narrow baseline window is imposed.

Select the primary follow-up within the inclusive interval from the index date plus 10 calendar months to the index date plus 14 calendar months. Choose the valid CDVA visit closest in absolute days to the 12-calendar-month anniversary; if equally close, choose the earlier visit. A visit outside this interval does not supply the primary endpoint. Preserve other dated visits for descriptive analysis. A nominal CSV column labelled 12 months is not sufficient to verify the window.

The parent text also mentions minimum follow-up of 12 months. The proposed clarification is to use its explicit 10–14-month endpoint window, allowing a valid visit before the anniversary. The clinical lead and research office must reconcile this wording with the approved protocol before primary analysis.

4 Outcomes and baseline predictors

Only predictors known before treatment are eligible. Do not feed postoperative refraction, haze, corneal measurements or follow-up availability into the model. Retain visual-acuity source notation and convert verified Snellen/decimal measurements according to the cleaning and transformation documents. UDVA never substitutes for CDVA; counting fingers/hand movements do not receive an invented logMAR value.

Corneal stability is reported from available longitudinal measurements and clinician documentation. The protocol does not create an unsupported progression threshold or call a single stable measurement proof of arrested disease.

Role	Prespecified definition
Primary outcome	Observed CDVA at the selected 10–14-month visit, in logMAR. Lower values indicate better vision.
Main predictors	Baseline CDVA (logMAR), age at treatment (years), recorded sex, verified treatment cohort, baseline Kmax (dioptres) and baseline thinnest corneal thickness (TCT, micrometres). TCT is the single prespecified pachymetry metric.
Clinical benchmark	Linear regression using baseline CDVA and treatment cohort, evaluated on the same eyes and splits as the main model. A training-set mean-only predictor supplies a reference.
Secondary continuous outcomes	UDVA, CDVA change, sphere, cylinder, spherical equivalent, K1, K2, Kmax and TCT at the selected visit where available; report paired denominators for changes.

Role	Prespecified definition
Recorded safety outcomes	Documented haze and its grade, delayed epithelial healing, infection, retreatment and other complications. Unknown documentation remains unknown. Absence of a complication cannot be inferred from an empty field.
Exploratory scope	Other follow-up visits and clinical stability descriptions are descriptive. Haze resolution is not analysed as time-to-event without verified onset/resolution dates and censoring information.

5 Data collection and quality control

Authorized reviewers extract EHR and Pentacam measurements using the accompanying collection sheet and exact variable mapping. Keep an unchanged raw snapshot, a coded analytical file, a screening log and a correction/adjudication log. Verify eligibility, surgery details, patient/eye linkage, baseline CDVA and the primary outcome against source records. Resolve conflicting units, duplicate encounters and transcription concerns without replacing unusual valid observations merely because they are extreme.

Report missingness separately by variable and treatment cohort, including primary endpoint availability and reasons where documented. Missing primary outcomes are not imputed or replaced by last observation carried forward. Outcome-observed and outcome-missing eyes are compared descriptively on available baseline characteristics to assess potential follow-up selection bias.

6 Modelling, evaluation and sample adequacy

The main model is ridge-penalized linear regression with the six prespecified predictor terms and an unpenalized intercept. Do not add interactions, polynomial terms, automated feature selection or a competition among high-capacity models to this small cohort. Preprocessing and penalty selection are learned inside training partitions only. The separate statistical plan fixes the model, benchmarks, resampling, uncertainty calculations and reporting.

Internal evaluation uses repeated nested cross-validation split by patient, preventing fellow-eye leakage. Report MAE as the primary performance metric, plus RMSE, calibration intercept/slope and uncertainty. Any performance measure from a fit evaluated on its own training records is labelled apparent, not internally validated. Repeated folds are correlated and do not create additional patients.

A biostatistician assesses adequacy using the number of independent patients, observed outcomes, predictor parameters, missingness, outcome variation and continuous-outcome sample-size criteria described by Riley et al. [2]. Document anticipated R-squared and precision assumptions before fitting, using external evidence where available. The planned 60 eyes is not an automatic adequacy threshold. If the required analysis cannot be supported, deliver the data audit and descriptive cohort results, and report model development as not executed.

7 Current extract and execution readiness

The current root CSV is an unverified preliminary extract. The initial audit describes 79 eye-like records: 51 CXL/CXL-only, 27 PRK with CXL and one PTK with CXL. It reports only nine baseline corrected-vision entries and two nominal 12-month corrected-vision entries; one of the latter contains qualified Snellen notation requiring adjudication. It lacks patient identifiers and dates needed for linkage and endpoint verification. These counts are raw availability counts, not confirmed eligible cohorts or a model sample; repeat and retain the audit on the locked source snapshot.

Primary model development is not feasible on this extract. Obtain the authorized coded patient/eye export, dates, verified procedure definitions, source-checked baseline/endpoint CDVA and baseline predictors. Do not manufacture outcomes, infer patients, or switch the target to a more complete column and present it as the planned study.

Gate	Evidence required before execution
Governance	Recorded AI-extension IRB amendment or formal determination; authorized extraction/access and agreed protocol wording.
Identity and eligibility	Verified coded patient/eye identifiers, dated index procedure and source-adjudicated eligibility/cohorts.
Endpoint	Source-checked baseline CDVA and sufficient observed 10–14-month CDVA; valid units and outcome variation.
Analysis feasibility	Biostatistician documents sample adequacy, feasible grouped splits, predictor variation and missingness strategy.
Release	Freeze source snapshot, dictionary, transformations, analysis plan and software manifest before fitting; record subsequent amendments.

8 Governance, public data and limitations

The revised idea PDF describes the parent study as IRB-approved and requires an amendment or formal determination for the AI extension before execution. Verify the approval reference, applicable consent/waiver and allowed data use rather than treating the form's filename as proof of approval. The source IRB form contains both consent and waiver wording; use the IRB's actual determination. Coded identifiers stay in the analytic file; MRN and direct identifiers remain outside it, with access restricted to authorized staff and approved KKESH systems.

Public datasets may support a separately labelled image, diagnostic or software demonstration after licence and provenance checks. A keratoconus image/classification dataset does not supply a post-treatment 12-month CDVA target. External prognostic validation requires compatible baseline variables, verified treatment and longitudinal CDVA definitions, independent patients and an appropriate licence. Do not pool auxiliary images or synthetic records into the KKESH clinical outcome cohort.

Separate public-data feasibility option: the supporting workbook for Wajnsztajn et al. [4] provides CXL clinical data, with 613 records and 456 source patient IDs in the downloaded audit. Its nominal one-year logMAR field contains 336 entries; 334 have a paired baseline value across 251 source IDs before eligibility/quality exclusions. This enables assessment of a CXL-only prognostic prototype, separately from the KKESH study. It includes conventional and accelerated CXL, has no TG-PRK cohort, and lacks dates/laterality needed to verify the KKESH endpoint. Its source-reported central pachymetry is not TCT. Retain the licence/attribution, screen adults and unresolved duplicate/chronology flags, and label the target nominal one-year CDVA. This is a feasibility experiment, not external validation of the intended two-cohort model.

Expected limitations include convenience sampling, a small effective patient sample, missing follow-up, retrospective measurement inconsistency, treatment-selection confounding and single-centre generalizability. Penalization and internal validation reduce some risks but do not establish clinical safety, causal superiority or external validity. Reporting will follow TRIPOD+AI and design appraisal will use PROBAST+AI [1,3].

9 Delivery and references

Day 1: obtain authorized data, verify definitions and freeze the protocol. Day 2: perform the feasible prespecified analysis and validation. Day 3: clinical/biostatistical review and reproducible table/figure delivery. The committee package is a complete prospective analysis specification; all results remain unpopulated until evidence and readiness gates are satisfied.

[1] Collins GS et al. TRIPOD+AI statement. *BMJ*. 2024;385:e078378. <https://doi.org/10.1136/bmj-2023-078378>

[2] Riley RD et al. Calculating the sample size required for developing a clinical prediction model. *BMJ*. 2020;368:m441. <https://doi.org/10.1136/bmj.m441>

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[3] Moons KGM et al. PROBAST+AI. BMJ. 2025;388:e082505. <https://doi.org/10.1136/bmj-2024-082505>

[4] Wajnsztajn D et al. Predicting factors for the efficacy of cross-linking for keratoconus. PLoS One. 2022;17:e0263528. <https://doi.org/10.1371/journal.pone.0263528>

Source protocol: revised KeraNova Project Idea PDF and supplied parent IRB application. Accompanying documents 2–7 form the implementation specification.

